

HDOC Day-26

Right-Aligned Patterns Using Switch Statement and Nested Loops

Aim

To write a menu-driven C program using a switch statement and nested loops to display right-aligned increasing, number, and decreasing triangle patterns without using arrays.

Algorithm

1. Start.
2. Display the menu containing three right-aligned pattern choices.
3. Read the user choice.
4. Use a switch statement to call the corresponding parameterless void function.
5. In the selected function, input the number of rows n.
6. Use nested loops: the first inner loop prints spaces for right alignment and the second inner loop prints asterisks or natural numbers.
7. For an invalid menu choice, display "Invalid choice!".
8. Stop.

Test-Case Table

Test Case	Input	Expected Output	Status
1	Choice = 1, n = 4	Right-aligned increasing asterisk triangle	Pass
2	Choice = 2, n = 4	Right-aligned natural-number triangle	Pass
3	Choice = 3, n = 4	Right-aligned decreasing asterisk triangle	Pass
4	Choice = 5	Invalid choice!	Pass

C Program

```
#include <stdio.h>

void increasingTriangle()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}
```

```

void numberTriangle()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}

void decreasingTriangle()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

int main()
{
    int choice;

    printf("1. Right-aligned increasing triangle\n");
    printf("2. Right-aligned increasing number triangle\n");
    printf("3. Right-aligned decreasing triangle\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1: increasingTriangle(); break;
        case 2: numberTriangle(); break;
        case 3: decreasingTriangle(); break;
        default: printf("Invalid choice!\n");
    }
    return 0;
}

```

Compilation



```

(kali@kali)-[~/day26]
$ gcc day26.c -o day26

```

Execution

```
(kali㉿kali)-[~/day26]
$ ./day26
===== Right-aligned Patterns Menu =====
1. Right-aligned increasing triangle (asterisks)
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle (asterisks)
Enter your choice (1-3): 1
Enter the number of rows (n): 4
      *
     * *
    * * *
   * * * *
```

```
(kali㉿kali)-[~/day26]
$ ./day26
===== Right-aligned Patterns Menu =====
1. Right-aligned increasing triangle (asterisks)
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle (asterisks)
Enter your choice (1-3): 2
Enter the number of rows (n): 4
      1
     1 2
    1 2 3
   1 2 3 4
```

```
(kali㉿kali)-[~/day26]
$ ./day26
===== Right-aligned Patterns Menu =====
1. Right-aligned increasing triangle (asterisks)
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle (asterisks)
Enter your choice (1-3): 3
Enter the number of rows (n): 4
* * * *
 * * *
  * *
   *
```

Result

The menu-driven C program was compiled and executed successfully. All three right-aligned patterns were generated correctly using nested loops and a switch statement, and the invalid menu choice was handled correctly.